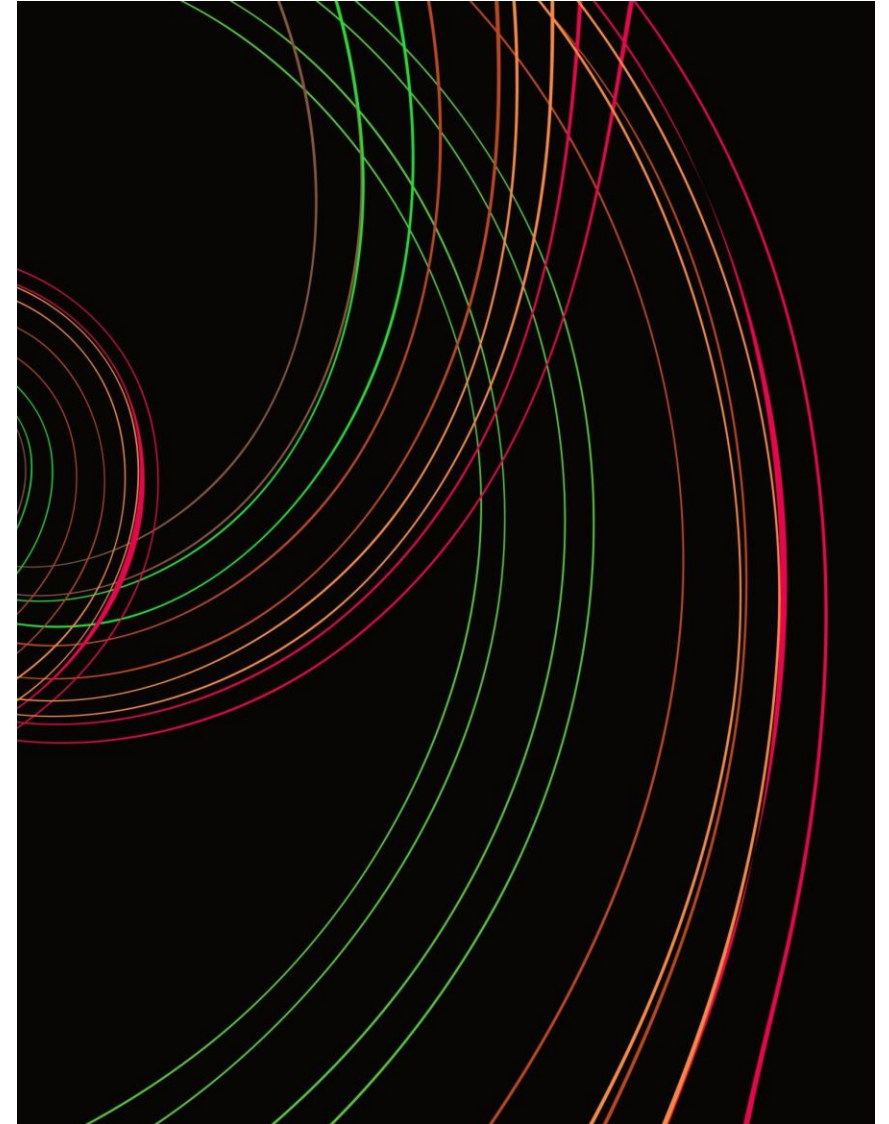


THE MEANING OF AISOC (AI-AUGMENTED SOC, AGENTIC SOC)

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WHO IS THE ATTACKER

A



Public Figure

B



Office Worker

C



Unknown Person

ANSWER - IDENTITY $\neq$ INTENT

Any of them or none.

- A: Office colleague $\rightarrow$ *Insider* can be negligent, compromised, or malicious.
- B: Unknown person $\rightarrow$ *External threat actor* (phishing, malware, hands-on-keyboard).
- C: Public Figure $\rightarrow$ Public Figure Used as Lure (impersonation target, not attacker).

Attacks ride identities and pretexts, not faces.

WHAT IS A SOC (SECURITY OPERATIONS CENTER)

- "A SOC is the computer safety team for a company."
- What they do
 - Watch - keep eyes on all computers and apps
 - Detect - spot weird or sneaky behavior.
 - Investigate - ask "what happened?"
 - Respond - fix it: block, reset, isolate
 - Improve - make better rules so it doesn't happen again
- Who they are:
 - Watchers (analysts), Fixers (incident responders),
 - Puzzle-solvers (threat intel), Builders (detection & automation)

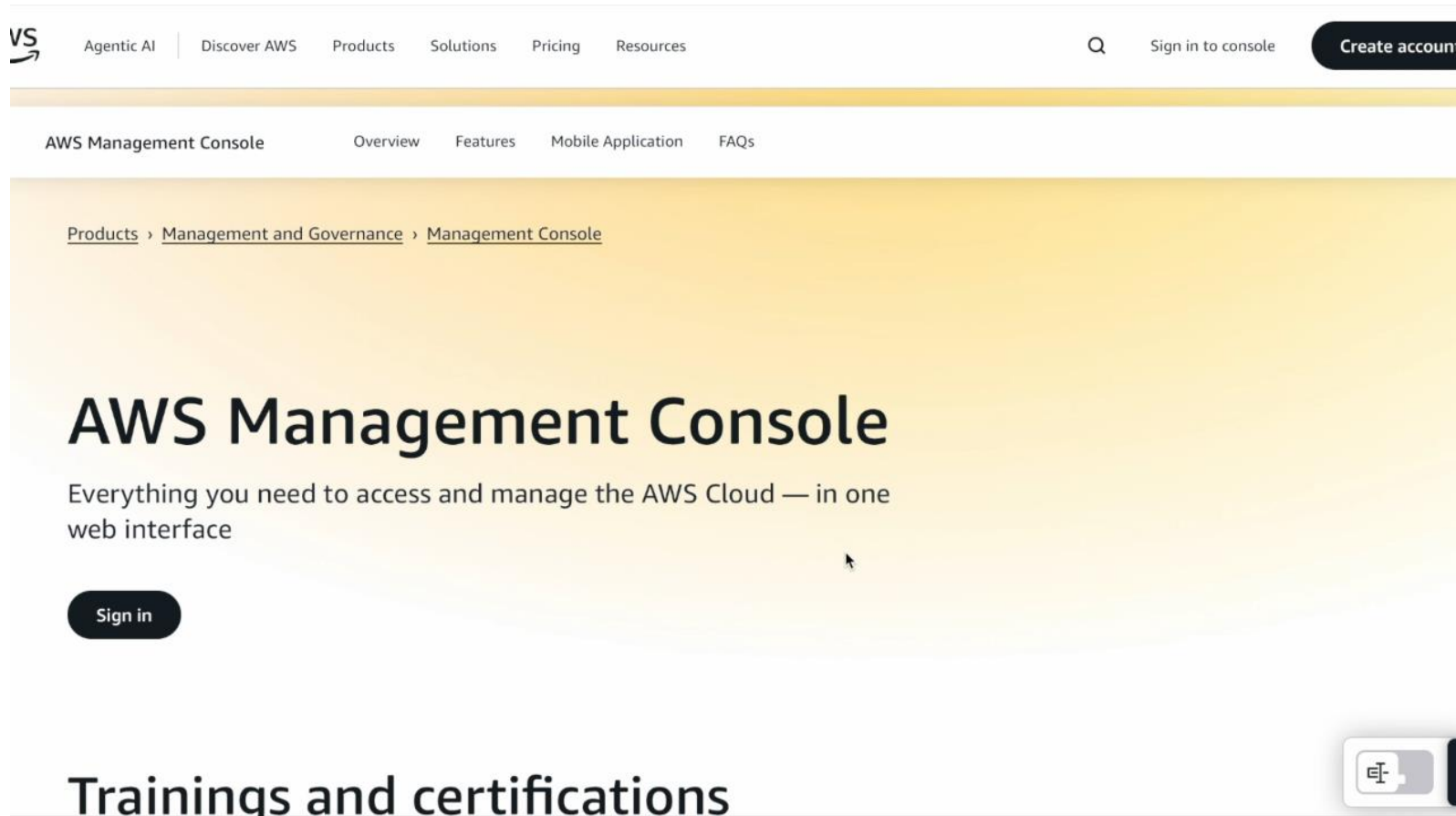
WHAT IS AN AI-SOC (AGENTIC $\neq$ AUTONOMOUS)

Agentic SOC: AI-assisted, task-oriented agents that help analysts *reason, triage, and act* – with human oversight.

Agent proposes $\rightarrow$ Analyst approves $\rightarrow$ SOAR executes $\rightarrow$ Audit log.

Not: "Fire-and-forget AI that runs your SOC."

AISOC IN PRACTICE



AISOC IN PRACTICE: MALWARE DETECTOR

Malware Model: 1.00 Precision, Recall, F1-score on validation and test sets

MALWARE MODEL PERFORMANCE ON VALIDATION AND TEST SPLITS.

Split	Precision	Recall	F1
Validation	1.00	1.00	1.00
Test	1.00	1.00	1.00

AISOC IN PRACTICE: LOG DETECTOR

Augmented Log Model: Cross-validation macro F1-score of 0.91 (after augmentation).

LOG MODEL ON FIXED VALIDATION AND TEST SPLITS (PROJECT LOGS).

Split	Setting	Precision	Recall	F1
Validation	Macro	0.75	0.93	0.79
Validation	Malicious	1.00	0.86	0.92
Validation	Benign	0.50	1.00	0.67
Test	Macro	0.58	0.64	0.59
Test	Malicious	0.92	0.79	0.85
Test	Benign	0.25	0.50	0.33

AISOC IN PRACTICE: FUSION

Final Fusion System: Macro F1-score of 1.00 across all alert levels on the held-out test set.

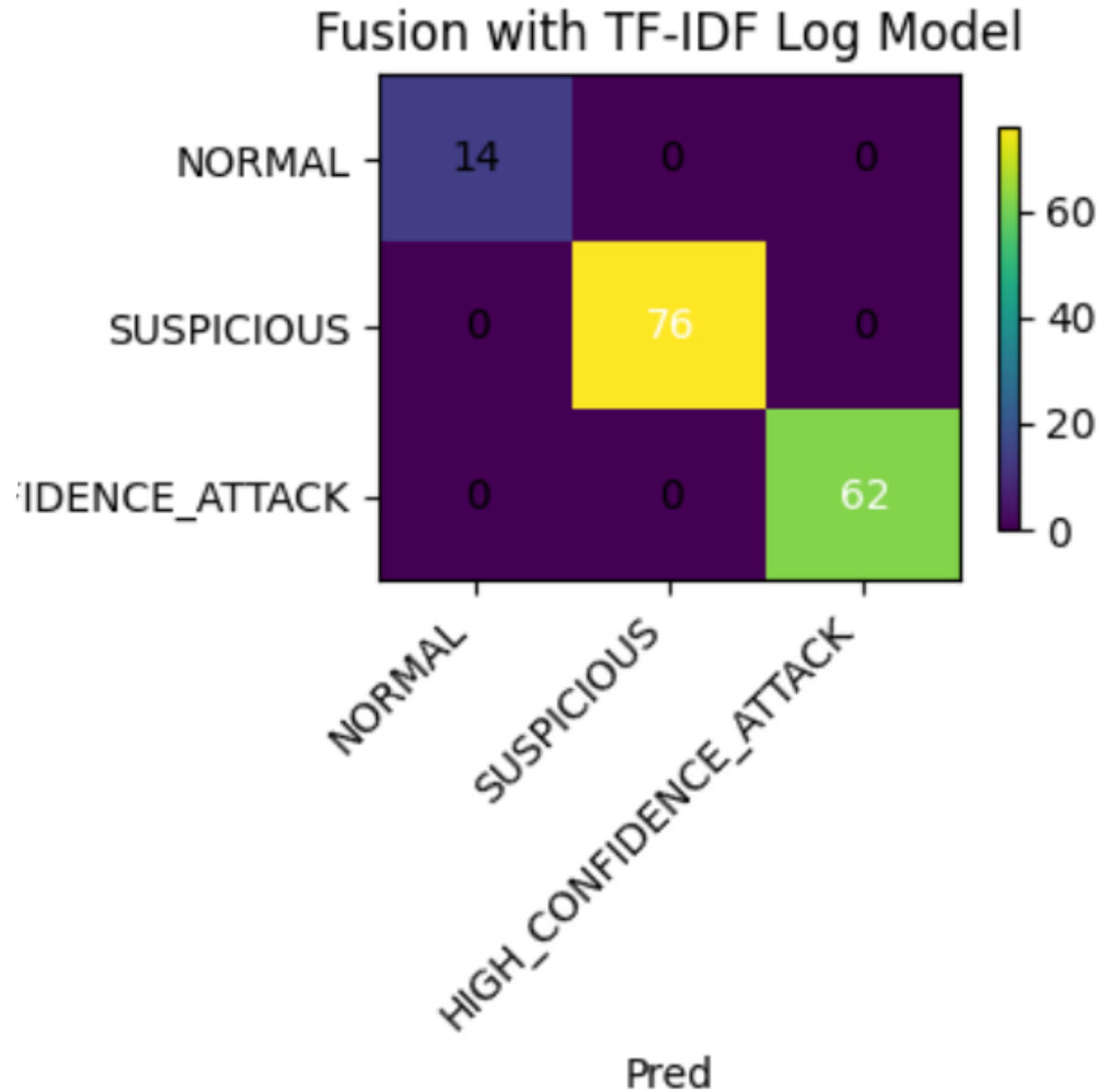
Controlled lab, held-out test – results won't generalize without noisy data & drift checks.

FUSED TRIAGE ON THE TEST SPLIT.

Class	Precision	Recall	F1	Support
NORMAL	1.00	1.00	1.00	14
SUSPICIOUS	1.00	1.00	1.00	76
HIGH_CONFIDENCE_ATTACK	1.00	1.00	1.00	62
Macro average	1.00	1.00	1.00	–

RESULT

- The results underscore the advantage of a multi-modal, AI-augmented approach in security operations. While individual models may suffer from data scarcity or limited scope, fusion mitigates these risks, delivering robust, high-confidence alerts. Data augmentation, especially for logs, proved critical to overcoming initial limitations and ensuring adequate model training. The success of the AISOC implementation highlights how AI can strengthen SOC operations against increasingly sophisticated cloud-based threats.



WHAT WORKED VS CAVEATS

What worked

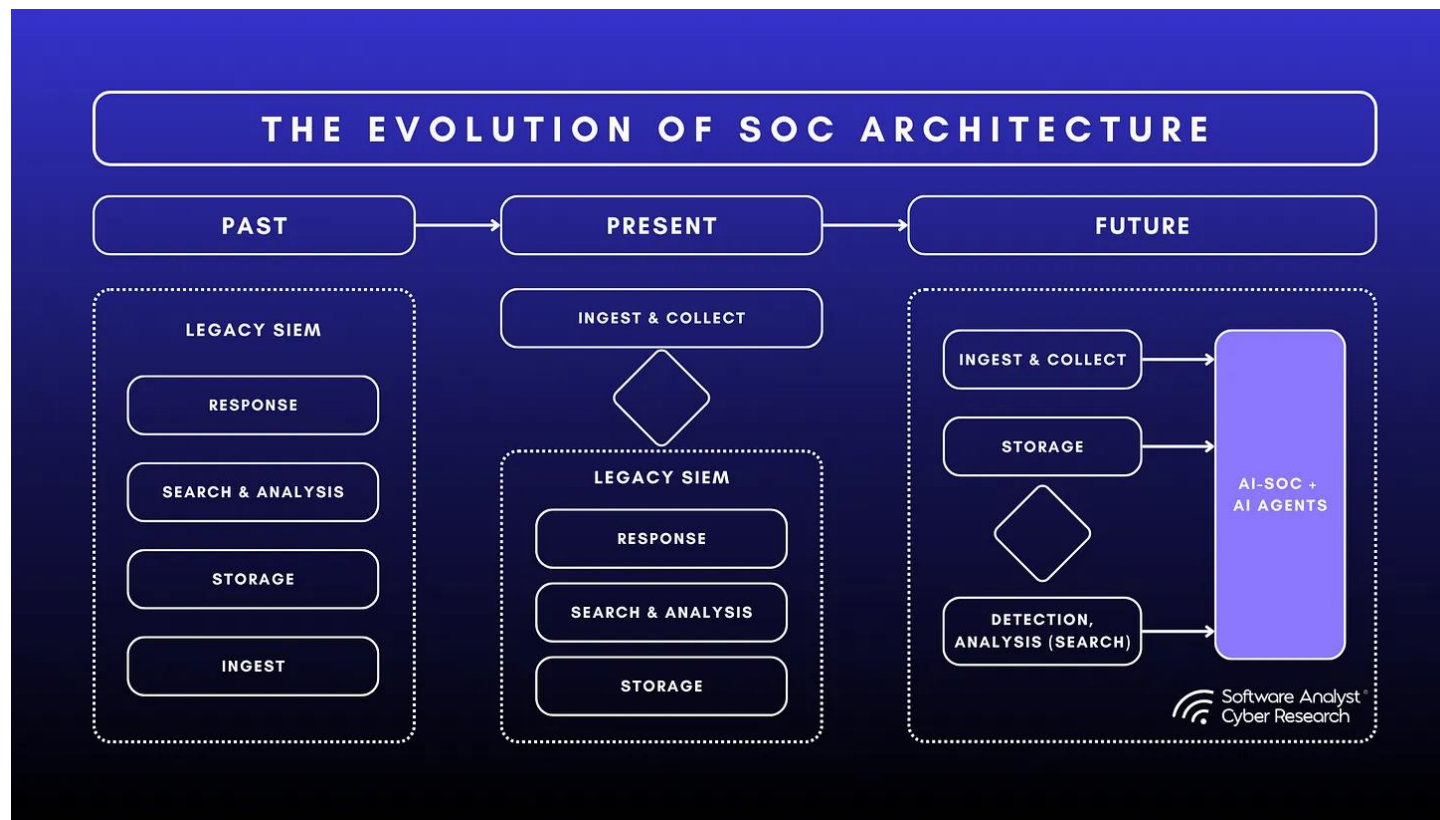
- Small, explainable models → fast & cheap.
- ELK + Filebeat → observability with minimal glue.
- Fusion rule → clear analyst mental model.

Caveats

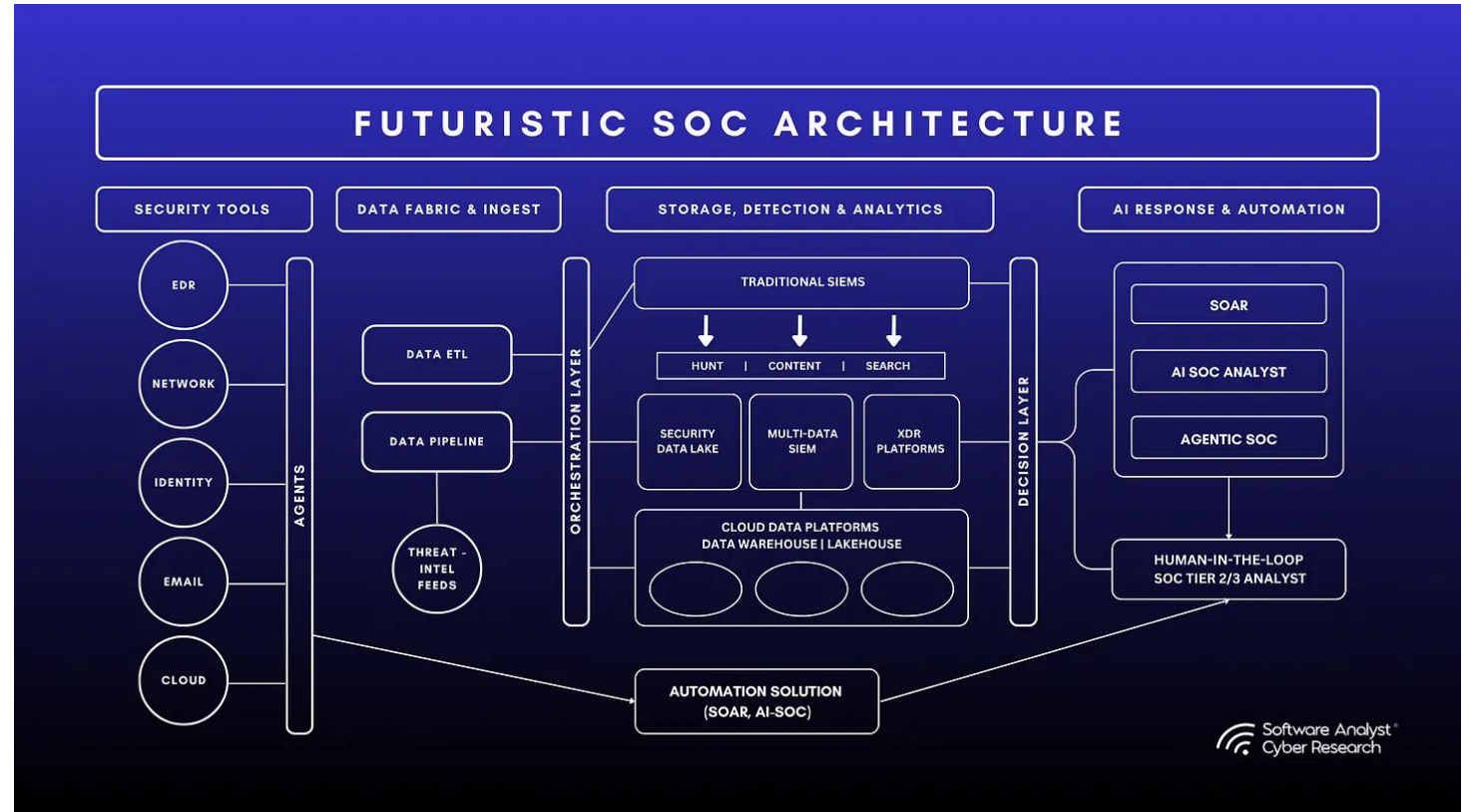
- Dataset shift & drift; adversarial obfuscation.
- Static malware features miss runtime tricks.
- LLM/agent reasoning can be wrong and confident.

EVOLUTION OF SOC ARCHITECTURE

- Reference: Software Analyst Cyber Research - <https://softwareanalyst.substack.com>



FUTURISTIC SOC ARCHITECTURE



- Reference: Software Analyst Cyber Research - <https://softwareanalyst.substack.com>



AI CAN REPLACE TIER-1 TODAY



SIMPLE FUSION CAN BEAT ONE
DETECTOR



HUMANS MUST APPROVE ACTIONS.

TWO TRUTHS AND A LIE

CONCLUSION

"AI is baked in-but analysts still run the show. That 's how you build a SOC that thinks with you, not over you" - CEO,
Underdefense, Nazar Tymoshyk